

Understanding Cascading Actions in MySQL

When we link two tables using a **Foreign Key**, the table containing the Primary Key is the **Parent Table**, and the table containing the Foreign Key is the **Child Table**.

By default, MySQL prevents you from deleting or updating a record in the Parent Table if it has related records in the Child Table. To change this behavior, we use **Cascading Actions**.

1. ON DELETE CASCADE

- **What it does:** If you delete a row in the Parent Table, MySQL will automatically delete all matching rows in the Child Table.
- **Use case:** If a customer deletes their account, you want all their past orders to be automatically deleted from the system.

2. ON UPDATE CASCADE

- **What it does:** If you change (update) the Primary Key value of a row in the Parent Table, MySQL will automatically update the matching Foreign Key values in the Child Table.
- **Use case:** If a department's ID number changes from 10 to 99, all employees belonging to that department will automatically have their department ID updated to 99.

Part: Practical Activity

Managing a system for **Vendors** (Suppliers) and their **Products**.

1. Create a database named **MarketDB**.
2. Create a parent table named **Vendors** with **VendorID** (INT, Primary Key) and **VendorName** (VARCHAR 50, Not Null).
3. Create a child table named **Products** with **ProductID** (INT, Primary Key), **ProductName** (VARCHAR 50, Not Null), and **VendorID** (INT).
4. Add a **Foreign Key** to the **Products** table that references **Vendors(VendorID)**. Apply **ON UPDATE CASCADE** and **ON DELETE CASCADE** to this relationship.
5. Insert the following Vendors:
 - VendorID: 1, Name: 'TechCorp'
 - VendorID: 2, Name: 'Foodies'
6. Insert the following Products:
 - ProductID: 101, Name: 'Laptop', VendorID: 1
 - ProductID: 102, Name: 'Mouse', VendorID: 1
 - ProductID: 103, Name: 'Cake', VendorID: 2
7. **Test Update Cascade:** Change the **VendorID** of 'TechCorp' from **1** to **99** in the **Vendors** table. Write a **SELECT** query to view the **Products** table and observe what happened to the Laptop and Mouse.
8. **Test Delete Cascade:** Delete the vendor 'Foodies' (VendorID 2) from the **Vendors** table. Write a **SELECT** query to view the **Products** table and observe what happened to the Cake.

```
-- Step 1: Create Database
CREATE DATABASE MarketDB;
USE MarketDB;
```

```

-- Step 2: Create Parent Table
CREATE TABLE Vendors (
    VendorID INT PRIMARY KEY,
    VendorName VARCHAR(50) NOT NULL
);

-- Step 3 & 4: Create Child Table with Cascading rules
CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50) NOT NULL,
    VendorID INT,
    FOREIGN KEY (VendorID) REFERENCES Vendors(VendorID)
    ON UPDATE CASCADE
    ON DELETE CASCADE
);

-- Step 5: Insert Parent Data
INSERT INTO Vendors (VendorID, VendorName) VALUES
(1, 'TechCorp'),
(2, 'Foodies');

-- Step 6: Insert Child Data
INSERT INTO Products (ProductID, ProductName, VendorID) VALUES
(101, 'Laptop', 1),
(102, 'Mouse', 1),
(103, 'Cake', 2);

-- =====
-- Step 7: Testing ON UPDATE CASCADE
-- =====
UPDATE Vendors
SET VendorID = 99
WHERE VendorID = 1;

-- If you run this SELECT query, you will see the VendorID for Laptop and Mouse has
-- automatically changed to 99!
SELECT * FROM Products;

-- =====
-- Step 8: Testing ON DELETE CASCADE
-- =====
DELETE FROM Vendors
WHERE VendorID = 2;

-- If you run this SELECT query, the 'Cake' product will be completely gone because
-- its Parent (Foodies) was deleted!
SELECT * FROM Products;

```